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ADVANCED MICROECONOMICS

Decisions under Risk and Uncertainty III - Insurance Markets

Winter Term 2025/2026

Outline

1) Consumer Theory

- 1.1) Preferences and Utility (L1)
- 1.2) Utility Maximization (L2)
- 1.3) Expenditure Minimization (L3)
- 1.4) Demand Functions and Duality (L4+L5)

2) General Equilibrium

- 2.1) Exchange Economy (L6)
- 2.2) Welfare Economics and General Equilibrium (L7)

3) Decisions under Risk and Uncertainty

- 3.1) Expected Utility (L8)
- 3.2) Risk Preferences (L9)
- 3.3) Insurances (L10)
- 3.4) Safety Regulations and the Value of a Statistical Life (11)

4) Behavioral Economics (L12)

Literature

- Autor, David (2016): Lecture Notes 17
- Planet Money Podcast - How hurricanes became a hot investment, December 6, 2025

Recap: Decisions under Risk and Uncertainty

- **Expected utility theory** gives us a powerful but stylized tool to model decision under risk - albeit using strong assumptions.
- Individuals decide between simple (reduced) lotteries so as to maximize their expected utility.
- Expected utility is the weighted sum of utilities in the different possible states of the world, where the weights are the probabilities that these states of the world are realized.
- Risk preferences are captured via the utility function:
 - risk averse – concave utility function
 - risk loving – convex utility function
 - risk neutral – linear utility function

Implications & Applications

- (vNM) Expected utility theory is an analytical machinery to model choices under risk, considering individual heterogeneity in preferences for risks via the utility functions
- But what for?
 - Using VNM expected utility, we can formally derive individuals' **willingness to pay** to avoid, reduce, or transfer risk.
 - This allows us to analyze
 - **explicit markets for risk** - insurance markets [*today*] and
 - **implicit markets for risk** that help us learn about the (statistical) value of life [*next week*]

Explicit Markets for Risk - Insurance

- Insurance markets are **explicit markets for risk**, i.e. markets on which risks are traded
- What is traded? - A certain payment, the premium, is exchanged for the promise to pay partial or full compensation (cover) for loss resulting from specified events
- Premium is exchanged for cover
- Insurance is a technology to **transfer risk from one person to another**
- The market for insurance is huge:
 - 428.8 mn individual contracts, 193.9 bn Euro revenue, 163.7 bn Euro paid out to cover losses in Germany (2019)
 - German market accounts for approximately 5% of the world market

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Demand for insurance - Setup

Assume an individual knows

- her **initial wealth** w_0 .
- her probability of having an accident p .
- the **reduction in wealth** (loss) associated with an accident L .

There are only **two possible states of the world**:

- 1 Accident (probability p)
- 2 No accident (probability $1 - p$).

The individual can buy an **actuarially fair** insurance.

Actuarially Fair

means that the insurance premium is equal to the expected insurance payout (the insurance breaks even in expected value): $p \times A$, where A is the amount of money the insurance pays in case of an accident.

Demand for insurance

Wealth with and without insurance in different states of the world:

	No accident ($1 - p$)	Accident (p)
Wealth (without insurance)	w_0	$w_0 - L$
Wealth (with insurance)	$w_0 - pA$	$w_0 - pA - L + A$

- Expected utility without insurance ($I=0$):

$$E(U|I = 0) = (1 - p)u(w_0) + pu(w_0 - L)$$

- Expected utility with insurance ($I=1$):

$$E(U|I = 1) = (1 - p)u(w_0 - pA) + pu(w_0 - L + A - pA)$$

Demand for insurance

How much insurance (i.e., what amount of A) does a **risk-averse individual** want to buy?

Expected utility theory:

$$\max_A E(U) = (1-p)u(w_0 - pA) + pu(w_0 - L + A - pA)$$

F.O.C

$$\begin{aligned} \frac{\partial E(U)}{\partial A} &= -p(1-p)\frac{\partial u(w_0 - pA)}{\partial A} + p(1-p)\frac{\partial u(w_0 - L + A - pA)}{\partial A} = 0 \\ \rightarrow \frac{\partial u(w_0 - pA)}{\partial A} &= \frac{\partial u(w_0 - L + A - pA)}{\partial A} \\ \rightarrow A &= L \end{aligned}$$

If the premium is actuarially fair, buying **full insurance** ($A = L$) is optimal for the individual.

Demand for insurance

Intuition

- FOC implies that individuals are best off if marginal utility of wealth is the same in all states of the world.
- Why? For a **risk averse consumer**, utility of average certain wealth is larger than average utility of (risky) wealth.
- With decreasing marginal utility (risk aversion) this condition holds if **wealth is equally distributed across all states of the world**.
- An actuarially fair premium allows transferring wealth from one state of the world to the other at no cost (i.e., without a reduction in expected value of wealth).
- A risk averse individual will transfer money to reach equal distribution across all states of the world (i.e., full insurance).
- A **risk loving consumer** would look for largest possible difference between states of the world and would thus choose no insurance ($A = 0$)

Insurance as a State-Contingent Commodity

- Insurance can be interpreted as a **state-contingent commodity**:
 - A good that is purchased **before** uncertainty is resolved
 - Pays out only if a specific **state of the world** occurs
- Here, the two states are:
 - **No accident** (“good” state)
 - **Accident** (“bad” state)
- Buying insurance means:
 - Paying a premium in *all* states
 - Receiving a payout only in the accident state
- Thus, insurance allows individuals to **transfer wealth across states of the world**

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How do insurance markets work?

- Why do we have insurance markets even if we assume that all people are risk averse?
 - Who will sell/provide insurance?
 - Three distinct mechanisms by which insurances can operate...
 - ① Risk Pooling
 - ② Risk Transfer
 - ③ Risk Spreading (*see Section 4.2 on non-idiosyncratic risks*)
 - Simple approach: Market is **competitive** and **production costs of insurance** are zero
- Perfect elastic supply of insurance cover at a fair premium

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Risk-Pooling

Risk Pooling

Insurer enters into insurance contracts with a number of distinct individuals and thus many **independent (idiosyncratic) risks are pooled** (i.e. thrown into one bucket)

- **Law of large numbers** makes total risk “disappear” as the variance of the expected average loss (over the pooled independent risks) converges to 0.
- If an insurance company insures many individuals they know with near certainty for how many of these individuals the risk will be realized and the insurance thus bears very little risk.
- Mechanism which is underlying most **private** insurance markets

Risk Pooling - Example

Example

- Assume the probability that your bike is stolen is 10% ($p = 0.1$)
- If your bike is stolen, you'll lose its entire value (€500).
- The expected value of the lottery (bike stolen yes/no) that you could insure against is $-(0.1 \times 500) = -50$ €
- If you are risk averse you'd be willing to pay at least €50 (the actuarially fair premium) for a full bike-insurance.
- If the bike insurance insures 1,000 people (each with 10% risk of having the bike stolen and a value of the bike of €500), they would get $1,000 \times €50 = €50,000$ in premiums and would have to pay (with near certainty) $1,000 \times €500 \times 0.1 = €50,000$.
- The 1,000 (risk averse) enrollees prefer this situation to no insurance: $u(450) > 0.9u(500) + 0.1u(0)$.

Risk-Pooling

Consider one insured individual.

- Loss occurs with probability p .
- Define the random variable of a **potential theft**:

$$X_i = \begin{cases} 1 & \text{Loss} \\ 0 & \text{No Loss} \end{cases}$$

- X_i follows a **Bernoulli distribution**.

$$\mathbb{E}[X_i] = p, \quad \text{Var}(X_i) = p(1 - p)$$

Risk-Pooling

Now consider n insured individuals.

- Each faces the same loss probability p .
- Risks are **independent**.
- Total number of losses $S_n = \sum_{i=1}^n X_i$ follow a **Binomial distribution** with parameters (n, p)

$$S_n \sim \text{Binomial}(n, p)$$

$$\mathbb{E}[S_n] = np$$

$$\text{Var}(S_n) = np(1 - p)$$

Risk-Pooling

- To judge the risk that the insurance bears, we'll look at the **variance** in the **loss share**, i.e. fraction of insured individuals who experience a loss
- Define the loss share:

$$\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i = \frac{S_n}{n}$$

- Expected Value of the loss share

$$\mathbb{E}[\bar{X}_n] = \mathbb{E}\left[\frac{S_n}{n}\right] = \frac{1}{n}\mathbb{E}[S_n] = p$$

- The expected loss share does not depend on $n \rightarrow$ Pooling does not change expected losses.

Risk-Pooling

- **Variance** of the loss share

$$\text{Var}(\bar{X}_n) = \text{Var}\left(\frac{S_n}{n}\right) = \frac{1}{n^2} \text{Var}(S_n)$$

- Substitute $\text{Var}(S_n) = np(1-p)$:

$$\text{Var}(\bar{X}_n) = \frac{p(1-p)}{n}$$

- As $n \rightarrow \infty$:

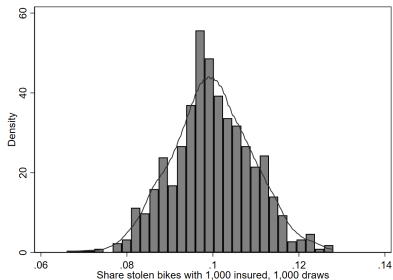
$$\text{Var}(\bar{X}_n) \rightarrow 0$$

- The loss share becomes nearly certain.
- Individual risk remains unchanged but fluctuations average out across many independent risks.
- Insurers face almost no aggregate uncertainty.

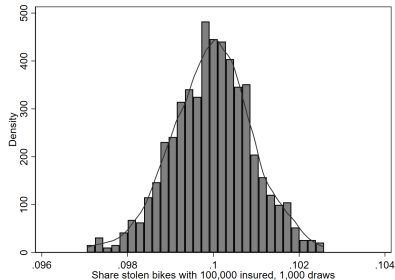
Risk-Pooling - Example

The larger n , the lower the variance of the loss share, i.e., the higher the certainty for the insurance ex ante of how many bikes will be stolen.

1,000 insured bike-theft risks



100,000 insured bike-theft risks



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Risk Transfer

- Risks can also be insured against (privately) by **transferring** them between individuals.
- This works as long as different individuals have **different degrees of risk aversion**.
- So far we've looked at risk aversion w.r.t a specific lottery (certainty equivalent, risk premium).
- Or more generally, looked at the shape of the utility function (convex, concave, linear).

Arrow-Pratt Measure of Risk Aversion

- But what about the **degree of risk aversion**?
- This depends on the **curvature** of the utility function:

Arrow-Pratt Measure of Absolute Risk Aversion (ARA)

$$r_A(x) = -\frac{u''(x)}{u'(x)}$$

Arrow-Pratt Measure of Relative Risk Aversion (RRA)

$$r_R(x) = -\frac{xu''(x)}{u'(x)}$$

- Intuition: The Arrow-Pratt measure captures how quickly marginal utility declines relative to its level.
- The higher the Arrow-Pratt index $-\frac{u''(x)}{u'(x)}$ is the higher the risk-aversion (*absolute or relative to x*)

Risk aversion and income

Constant Absolute Risk Aversion (CARA), e.g. $u(x) = -e^{-ax}$

$$r_A(x) = -\frac{u''(x)}{u'(x)} = -\frac{-a^2 e^{-ax}}{ae^{-ax}} = a$$

- With CARA, the risk aversion does not change with wealth (x).

Constant Relative Risk Aversion (CRRA), e.g. $u(x) = \ln(x)$

$$r_R(x) = -\frac{xu''(x)}{u'(x)} = -\frac{x(-x^{-2})}{x^{-1}} = 1$$

- Utility functions with CRRA are also called **iso-elastic utility functions**

CRRA and Risk Transfer

- CRRA implies **Decreasing Absolute Risk Aversion (DARA)**

$$r_A(x) = -\frac{u''(x)}{u'(x)} = \frac{1}{x}$$

- which means that the risk aversion of these individuals decreases with increasing wealth (x).
- If all individuals have a CRRA utility function then richer people will fear the same loss less than poorer people.
- Intuitively, everyone can thus be better off when trading risks between rich and poor (poor person pays rich person to take over the risk).

Risk Transfer - Example

- **Setting:**
 - 2 individuals with utility function $u(w) = \ln(w)$.
 - Individual 1 has initial wealth $w = 200,000$ and a 50/50-risk to lose 100,000.
 - Individual 2 has $w = 2,000,000$.
- What is the actuarially fair premium for full insurance?
$$\pi^{AF} = 0.5 \times 100,000 = 50,000$$
- What would individual 1 pay to get rid off the risk?
- And what about individual 2?

Risk Transfer

Example

- Individual 1:
 - Expected utility of lottery:

$$EU = 0.5 \ln(200,000) + 0.5 \ln(100,000) = 11.86$$

- Expected wealth: 150,000
- Certainty equivalent of lottery: $U(c) = 11.86$, i.e.,

$$CE = U^{-1}(11.86) = e^{11.86} = 141,421$$

- Risk premium: $150,000 - 141,421 = 8,579$.

Risk Transfer

Example

- Individual 2:
 - Expected utility of lottery:

$$EU(L) = 0.5\ln(2,000,000) + 0.5\ln(1,900,000) = 14.48$$

- Expected wealth: 1,950,000
- Certainty equivalent: $U(c) = 14.48$, i.e.,

$$CE = U^{-1}(14.48) = e^{14.48} = 1,949,359$$

- Risk premium: $1,950,000 - 1,949,359 = 641$.

Risk Transfer

Example

- Individuals could agree an **amount** of money that 1 would pay to 2 for 2 to take over 1's risk.
- **Who pays what?**

- Actuarially fair premium (expected loss):

$$\pi^{AF} = 0.5 \times 100,000 = 50,000.$$

- Individual 1's maximum total premium (willingness to pay):

$$\pi_{\max} = \pi^{AF} + RP_1 = 50,000 + 8,579 = 58,579.$$

- Individual 2's minimum total premium (willingness to accept):

$$\pi_{\min} = \pi^{AF} + RP_2 = 50,000 + 641 = 50,641.$$

- Mutually beneficial trade if:

$$50,641 \leq \pi \leq 58,579.$$

- Both individuals would be better off through this risk transfer (this is a pareto improvement).

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Asymmetric information

- In theory, insurance works very well: individuals are risk averse, all risks are known to everyone, and risks can therefore be pooled or transferred efficiently.
- However, the assumption that “*everyone knows all risks*” is not very realistic.
- In particular, the problem of **asymmetric information** is highly relevant in insurance markets, because
 - at the time of signing the insurance contract, the buyer typically knows more about her risk than the insurer.
 - after the contract is signed, the insurer cannot fully observe the behaviour of the insured.
- This leads to
 - adverse selection
 - moral hazard

Adverse selection

Private information before contract signing

- Extension of the simple model of insurance demand:
 - Individuals differ in their risk types.
 - Buyers know their own risk better than the insurer (private information).
- The insurer sets an actuarially fair premium for the average individual.
- This premium may be too high for low-risk individuals, who then choose not to buy insurance.
- As a result, only bad (high-risk) types remain insured.
- Private information leads to an inefficient outcome, because low-risk individuals cannot obtain insurance at a fair price.

Moral hazard

Private information after contract signing

- Insurers cannot (or can only imperfectly) observe individuals' behaviour after the contract is signed.
- Because insured individuals are no longer fully affected by losses (under full insurance), they may change their behaviour:
 - **ex ante moral hazard:** before a loss occurs, individuals take more risk.
 - **ex post moral hazard:** after a loss occurs, individuals use more expensive means of loss recovery.
- These behavioural changes increase the cost of insurance (because losses occur more frequently or generate higher costs), to the detriment of all insured individuals.
- This is, for example, a reason for **deductibles and co-payments** in insurance contracts.

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Correlated risks

- **Standard insurance** relies on risk pooling and diversification
- This works well for **independent risks** (i.e. risks that are independent of each other)
- Some risks involve **very large and highly correlated losses**:
 - Natural disasters (earthquakes, floods, hurricanes)
 - Terrorism, pandemics
- These risks are **not fully diversifiable** within a single insurer
- Insurers face **limited risk-bearing capacity** due to capital constraints
- Result: very high premiums, coverage limits, or missing markets

Risk Spreading

- Risks can be **spread** as long as there are individuals who are hit more and individuals who are hit less.
 - There are **10,000 people** with initial wealth $w_0 = 100,000$.
 - They all have the same utility function $u(w) = \ln(w)$.
 - If one of them is hit by a shock that leads to a loss of 50,000, the individual loses

$$u(100,000) - u(50,000) = 0.693$$

- If instead the loss is equally spread across all individuals, everyone had a loss of 5 and thus the total loss would be

$$10,000 \times (\ln(100,000) - \ln(99,995)) = 0.500$$

Solution: Government Intervention

- **Risk Spreading:** Risks could be **spread** as long as there are individuals who are hit more and individuals who are hit less.
- This is typically done through **government intervention** (transfers) in case of natural disasters that make society better off overall (increase the social welfare function).
- Government provides an implicit insurance against correlated risks.
- Risk spreading doesn't work on private market (as those who expect to be less affected won't participate).
- Thus this is a classical “market failure”, where the government can improve overall wellbeing by spreading risks (often ex post).

Risk Spreading - Example

- There are **10,000 people** with initial wealth $w_0 = 100,000$.
- They all have the same utility function $u(w) = \ln(w)$.
- If one of them is hit by a shock that leads to a loss of 50,000, he'll lose

$$u(100,000) - u(50,000) = 0.693$$

- If instead the loss is equally spread across all individuals, everyone had a loss of 5 and thus the total loss would be

$$10,000 \times (\ln(100,000) - \ln(99,995)) = 0.500$$

- **Welfare:** Assuming a **utilitarian social welfare function**, risk spreading would thus increase overall welfare.

Private Solutions

- **Reinsurance**
 - Insurers insure themselves against large losses
 - Risk is spread across multiple insurers (often globally)
 - Increases risk-bearing capacity of the insurance sector
- **Catastrophe bonds (Cat bonds)**
 - Insurance-linked securities sold to **capital market** investors
 - Investors receive high returns in normal times
 - If a catastrophe occurs: Investors bear the risk instead of insurers
 - Attractive to investors because less strongly correlated with standard capital markets risks
 - *See: Planet Money Podcast - How hurricanes became a hot investment, December 6, 2025*

Jamaica's Cat Bond

Bloomberg

• Live TV Markets Economics Industries Tech Politics Businessweek Opinion More

Hurricane Season: Hurricane Melissa | Supercharged Hurricanes | Deadly Tornadoes | Chaos Inside FEMA

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ESG & Investing

Jamaica Catastrophe Bond Headed for Full Payout After Hurricane, World Bank Says



Cars and damaged property in a flooded section of road from Holland Bamboo to Middle Quarters in St Elizabeth, Jamaica, on Oct. 31. Photographer: Ricardo Makyn/AFP/Getty Images

Source:

<https://www.bloomberg.com/news/articles/2025-11-07/world-bank-says-jamaica-cat-bond-headed-for-full-payout>
(Accessed December 16, 2025)

Summary: Insurance and Risk

- Expected utility theory allows us to analyze individuals' willingness to pay to avoid, transfer, or bear risk.
- With actuarially fair insurance, risk-averse individuals optimally choose full insurance.
- Insurance can make everyone better off (or at least increase social welfare) if individuals are risk averse.
- Insurance can operate through three main mechanisms:
 - **Risk pooling**: independent risks are diversified via the law of large numbers.
 - **Risk transfer**: risks are shifted to individuals with lower marginal disutility of risk (e.g. wealthier individuals).
 - **Risk spreading**: correlated risks can be socially mitigated through redistribution (often via government).
- The Arrow–Pratt measure captures the degree of risk aversion and explains why risk transfer can generate Pareto improvements.
- In practice, insurance markets face limitations due to asymmetric

Keywords

- Actuarially fair insurance, full insurance
- State-contingent commodities
- Risk pooling, law of large numbers
- Arrow–Pratt measure (ARA, RRA), CARA, CRRA
- Risk transfer, risk spreading
- Adverse selection, moral hazard
- Correlated risks, reinsurance, catastrophe bonds

Questions?

